

BATR Performance Analysis & Operational Improvement Plan

Data Analysis Project

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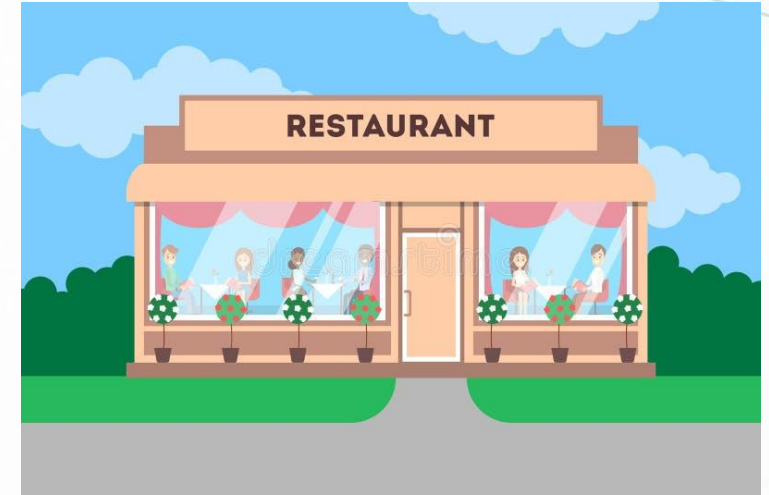
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Biker Arrival at Restaurant Time (BART)



Biker Arrival at Restaurant Time (BART)

Biker Acceptance Time (BAT)

Biker Post-Acceptance Arrival Time (BPAT)

What Is the Problem?

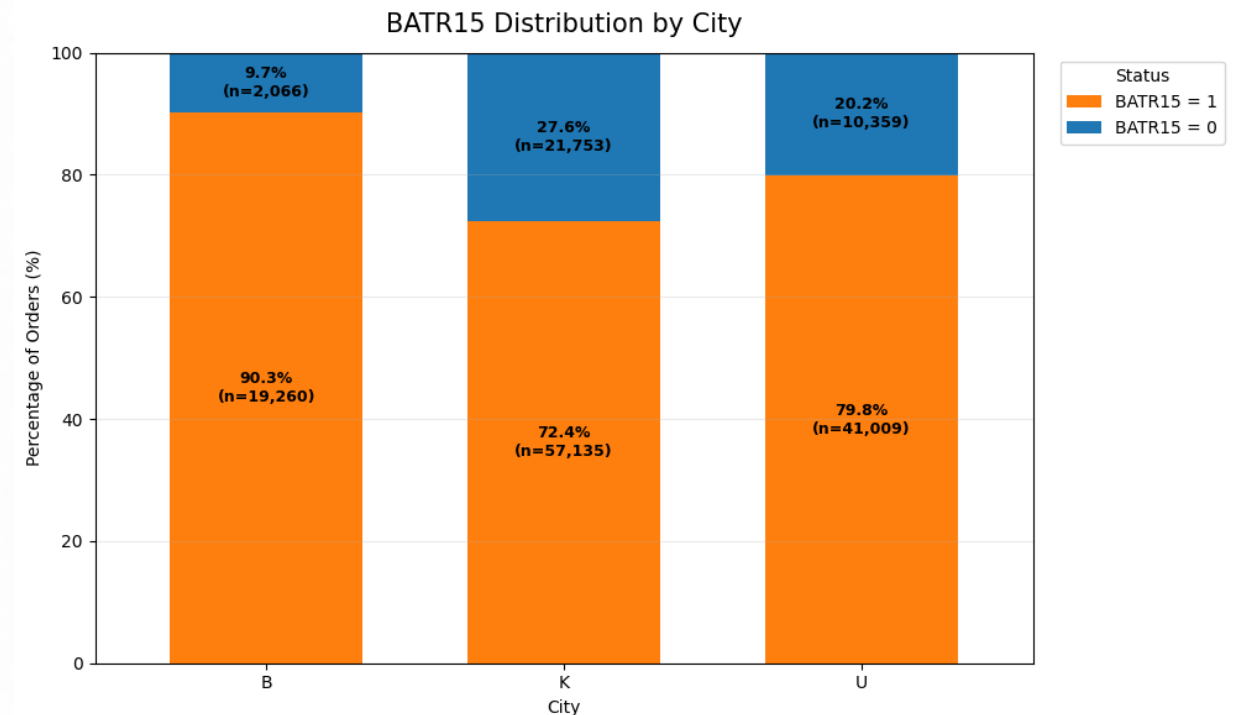
ATR performance is below the expected operational target, with overall **BATR15 at 77.45%** versus the **90% target**.

The key challenge is that poor BATR performance can come from different parts of the process.

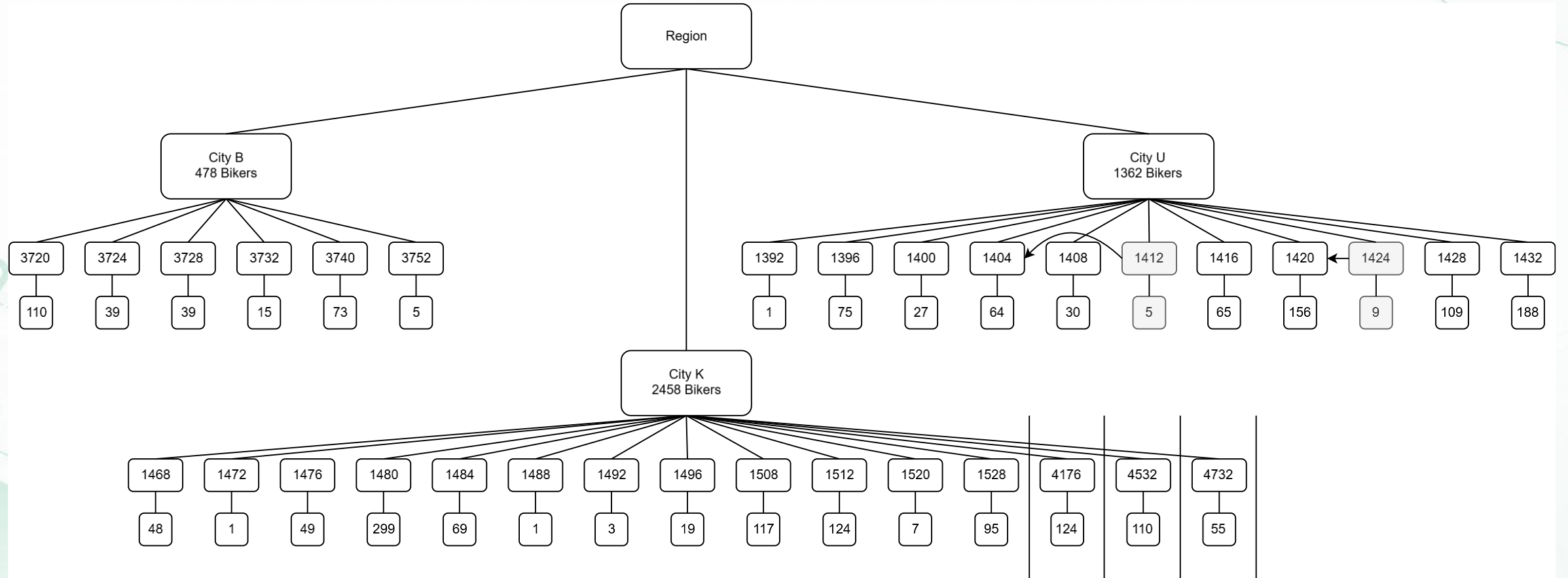
Adelay may be caused by a

long Acceptance Time, a long Arrival Time, or both.

BATR can also vary significantly across different locations and time periods, with some cities, districts, and peak hours performing much worse than others.



Region Structure

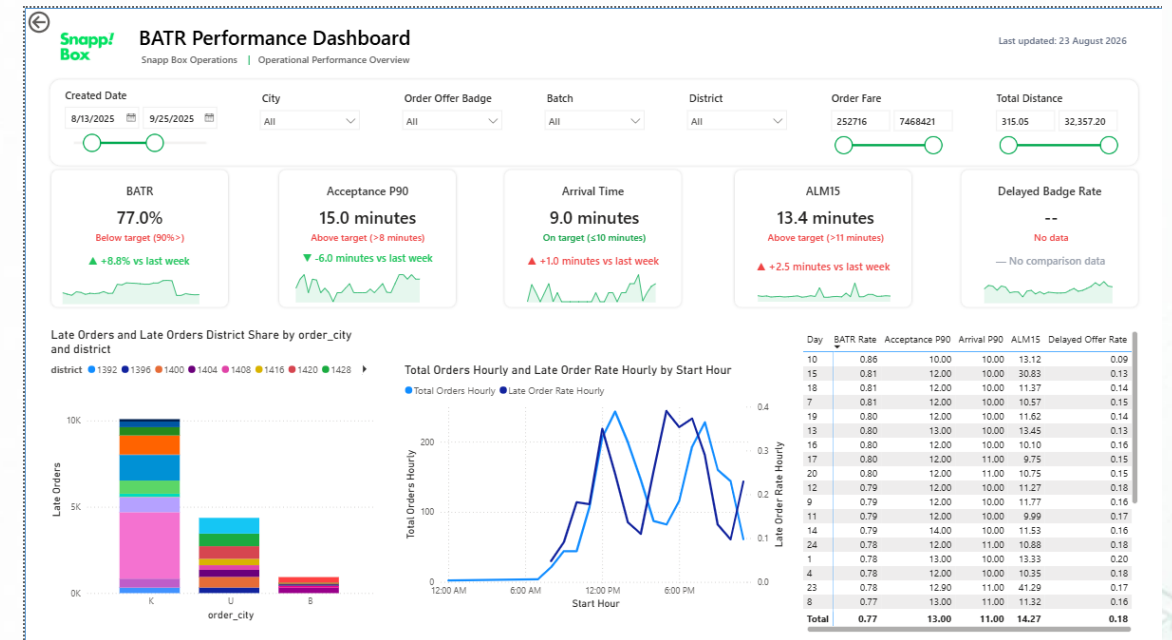


How We Monitor BATR Performance?

We monitor BATR through one Outcome KPI and four supporting operational indicators. Together, they show not only whether performance is deteriorating, but also where the deterioration is coming from.

Core KPIs

- **BATR15:** % of orders reaching the restaurant within 15 minutes
- **ALM15:** severity of delay among late orders
- **Acceptance P90:** tail performance before acceptance
- **Delayed Offer Rate:** early warning for the offer/acceptance process
- **Arrival P90:** tail performance after acceptance



Where Is the Largest Performance Gap?

The first major difference appears at the city level.

City B: BATR15 $\approx$ 90.3% — approximately on target

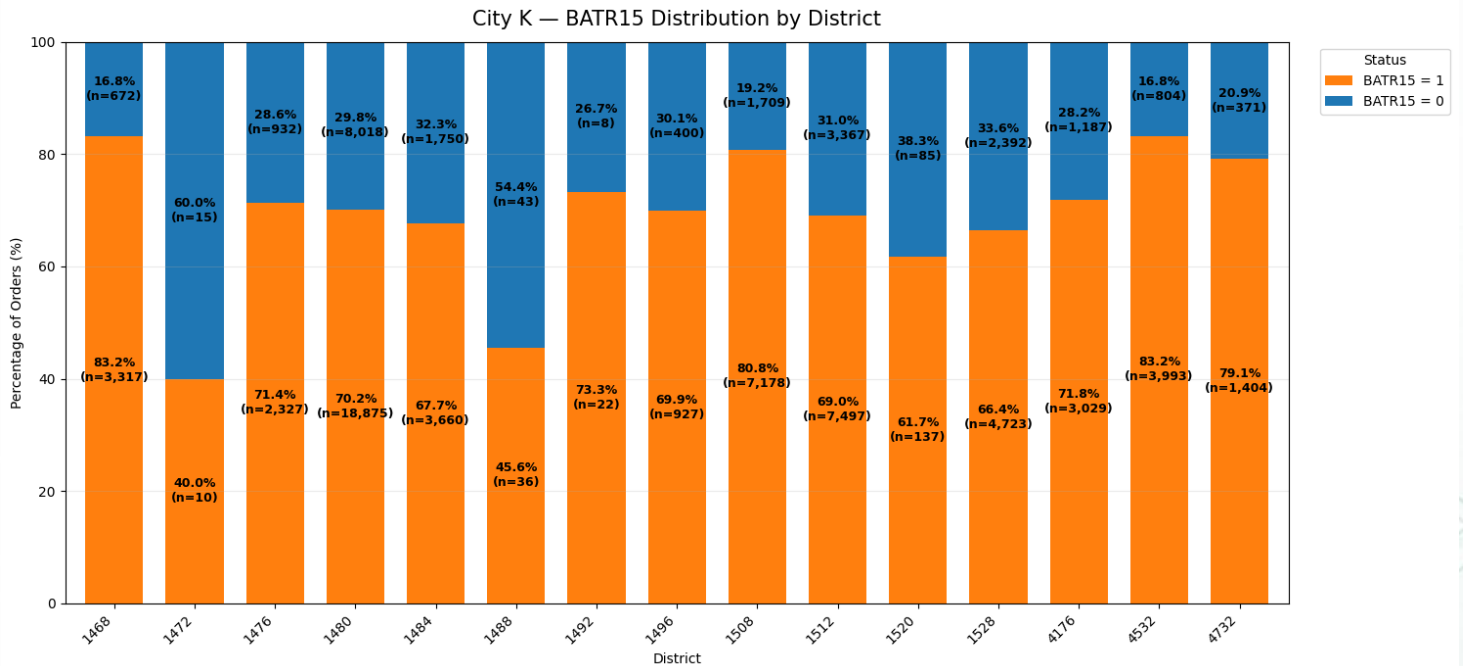
City U: BATR15 $\approx$ 79.8% — below target

City K: BATR15 $\approx$ 72.4% — weakest performance

City K is especially

important because it
generates approximately
63.6% of all Late Orders.

Therefore, improving
City K provides the largest
opportunity for
network-level impact.



Is the Problem Temporary or Persistent?

The three cities show very different trends.

City B gradually improved and exceeded the target in October.

City K remained weak throughout the three-month period, suggesting a persistent operational

issue rather than a

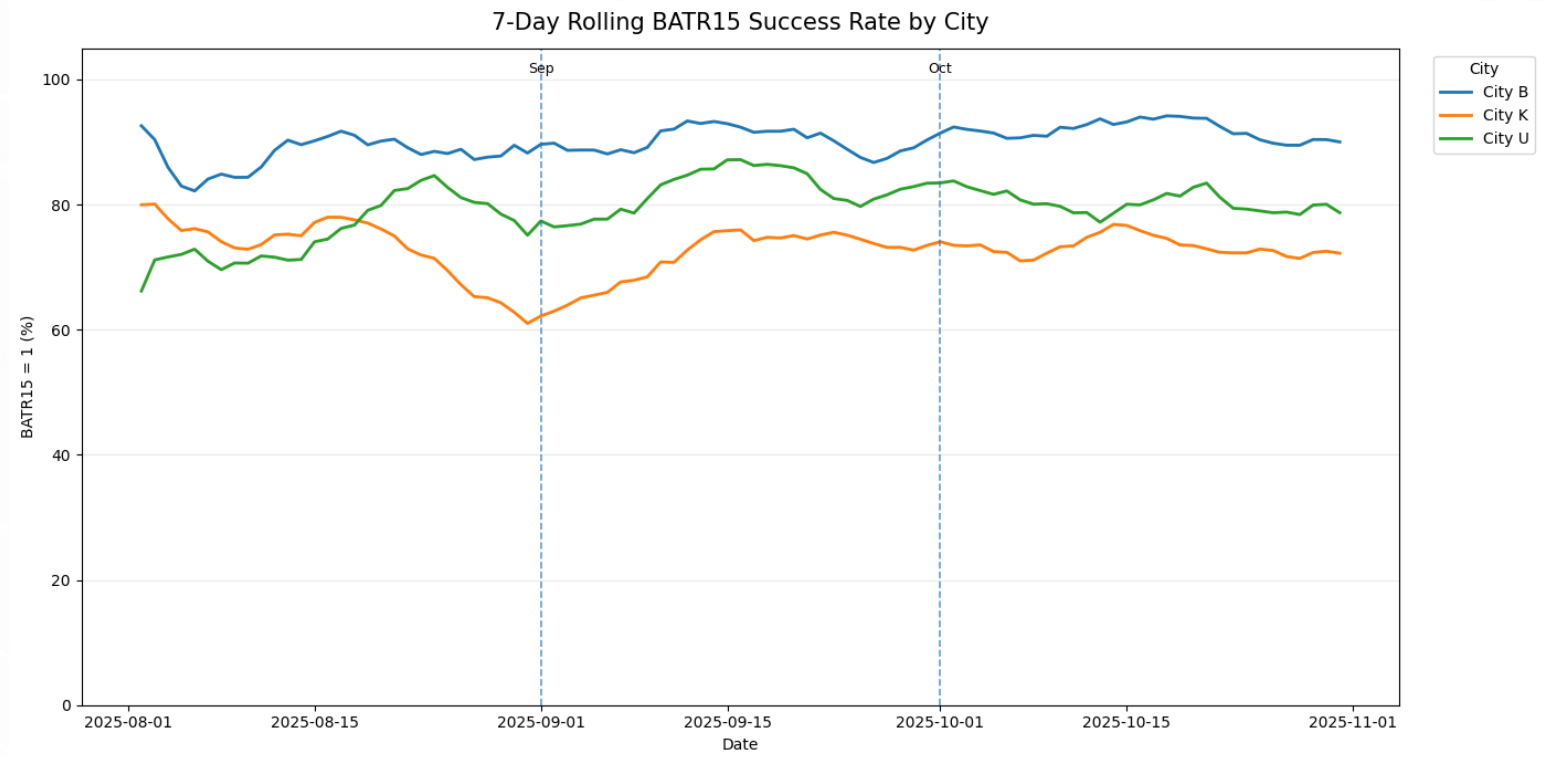
temporary shock.

City U improved in

September but

deteriorated again

in October.



Data Preparation and Data Quality

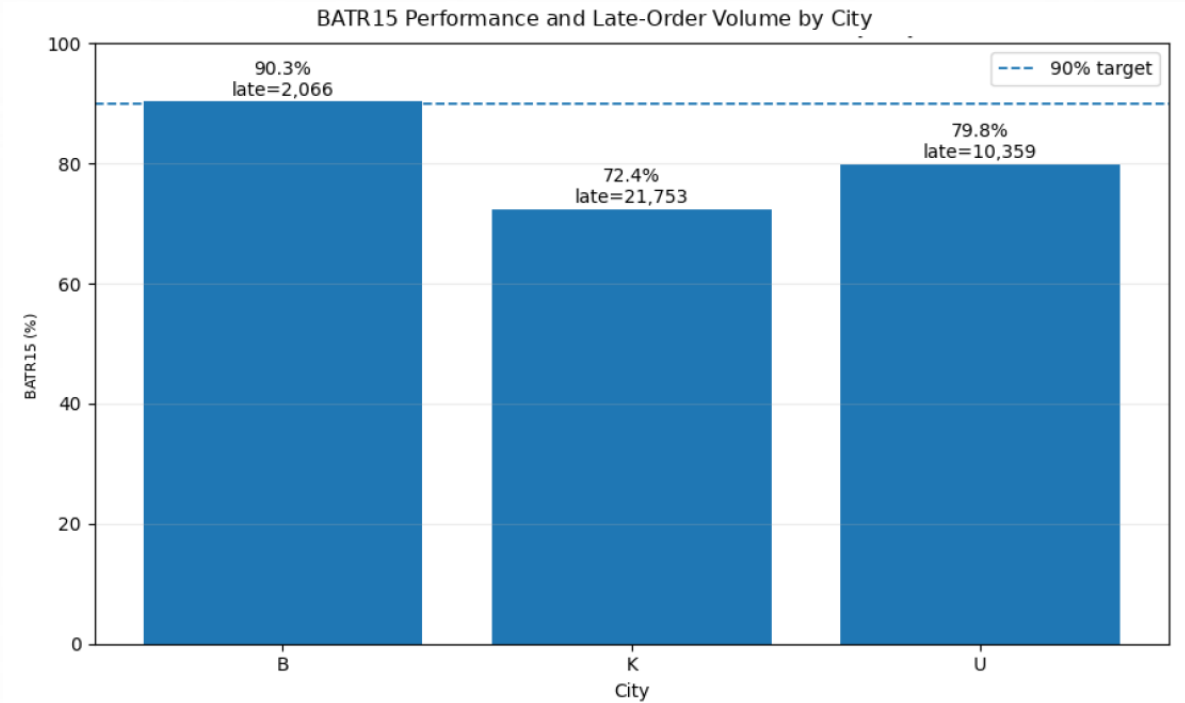
- 151,580 valid orders were analyzed, with an overall BATR15 of 77.45% versus the 90% target.
- 23,803 orders (15.7%) had inconsistent timestamps; these were retained for general KPIs but excluded from exact process decomposition.
- A Sequential Cohort of 127,774 orders (84.3%) was used for Acceptance vs. Arrival analysis.
- Small or temporary districts were treated cautiously to avoid misleading comparisons.

Data-quality item	Value
Merged orders	151803.0
Valid BATR orders Aug-Oct	151580.0
Sequential timestamp cohort	127774.0
Sequential cohort coverage	84.29
arrived_pickup_at before accept_time	23803.0
Missing start_time_slot	221.0
Missing accept_time	224.0
Missing biker	16.0

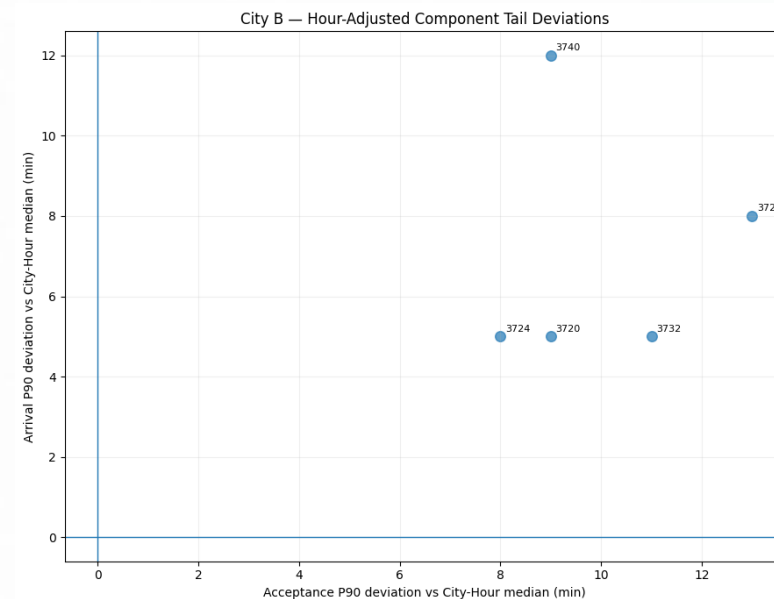
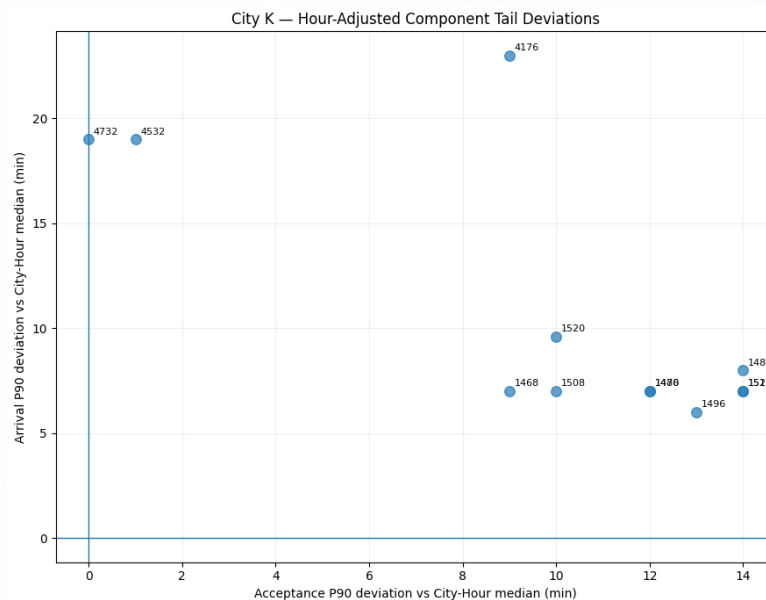
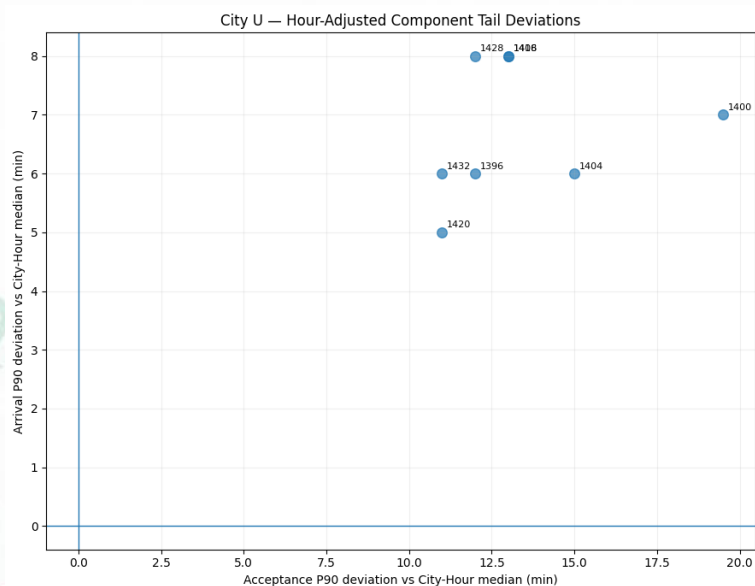
Overall and City-Level Performance

- City B meets the 90% target with 90.3% BATR15.
- City K is the weakest performer at 72.4% and generates 63.6% of all Late Orders.
- City U also remains below target at 79.8%.

City	Orders	BATR15	Late Orders	Share of All Late Orders
B	21,324	90.31%	2,066	6.0%
K	78,888	72.43%	21,753	63.6%
U	51,368	79.83%	10,359	30.3%

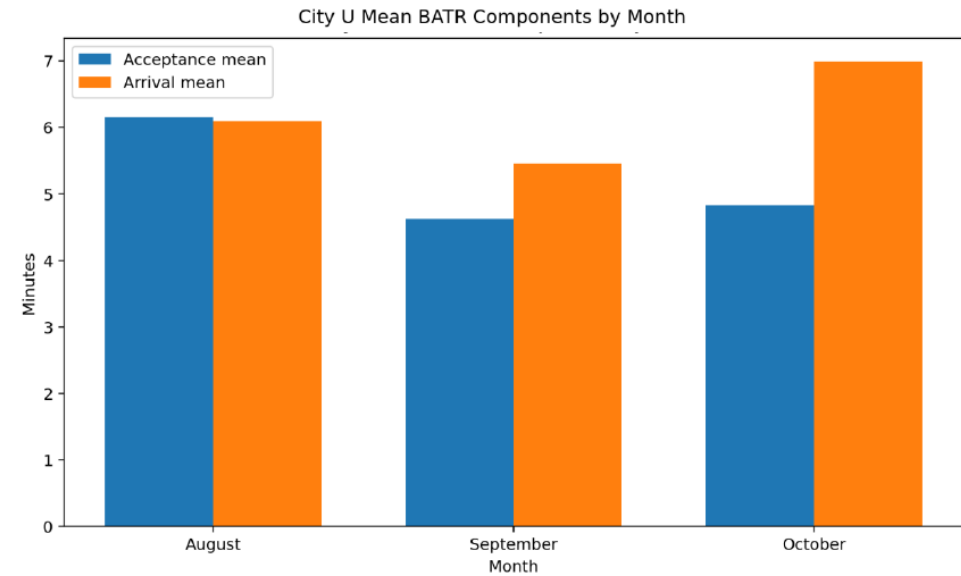
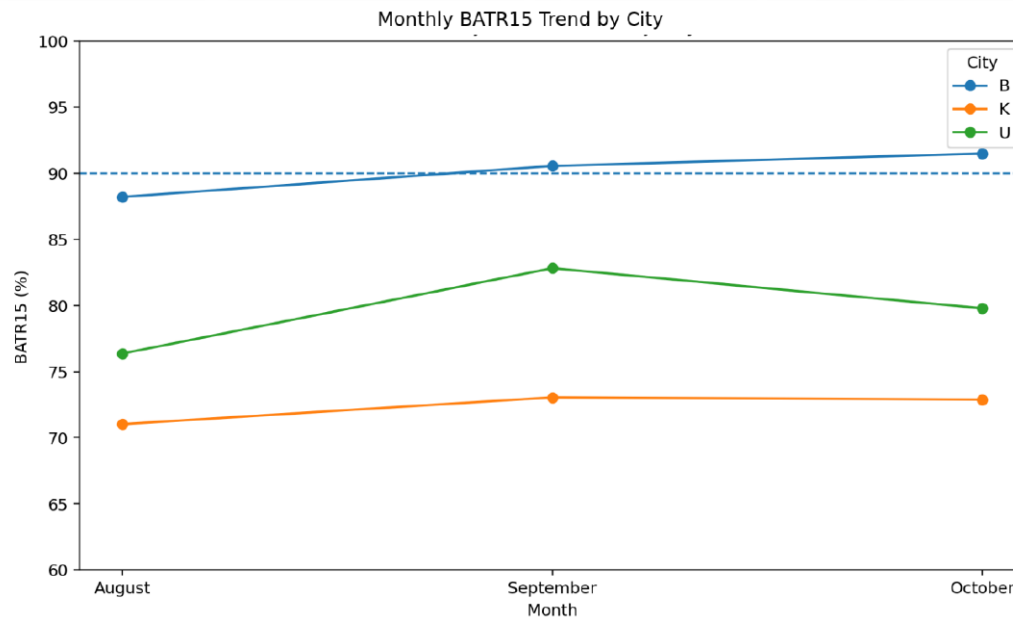


Not Every Low-Performing District Has the Same Problem



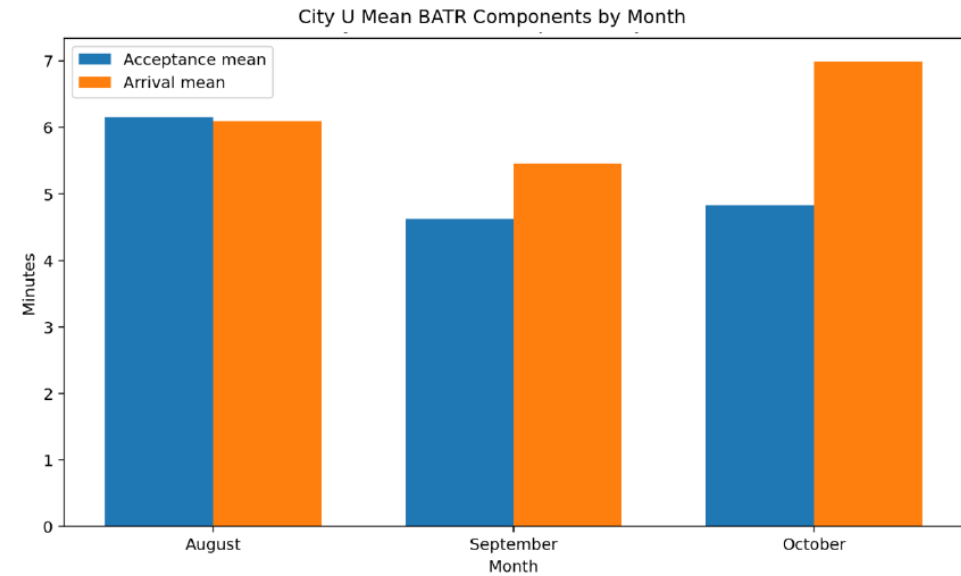
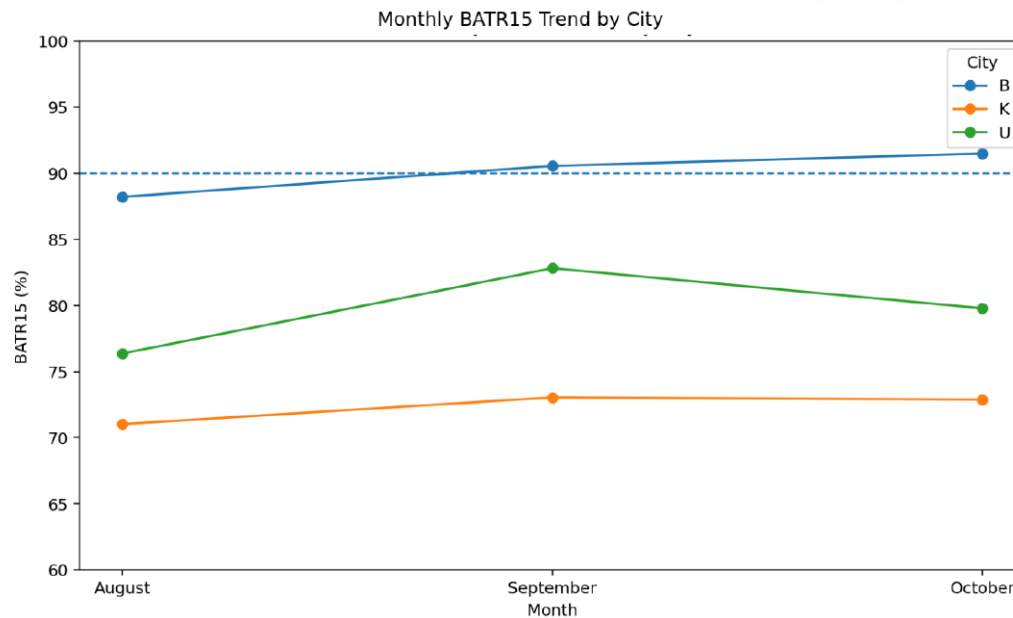
Performance Trend over Time

- Overall BATR15 improved from 75.2% in August to 78.8% in September, then slightly declined to 77.9% in October.
- City U: Mean BATR increased by 1.74 min from September to October. Only 0.21 min came from Acceptance, while 1.53 min came from Arrival.



Performance Trend over Time

- Overall BATR15 improved from 75.2% in August to 78.8% in September, then slightly declined to 77.9% in October.
- City U: Mean BATR increased by 1.74 min from September to October. Only 0.21 min came from Acceptance, while 1.53 min came from Arrival.



District-Level Performance and Context-Adjusted Benchmark

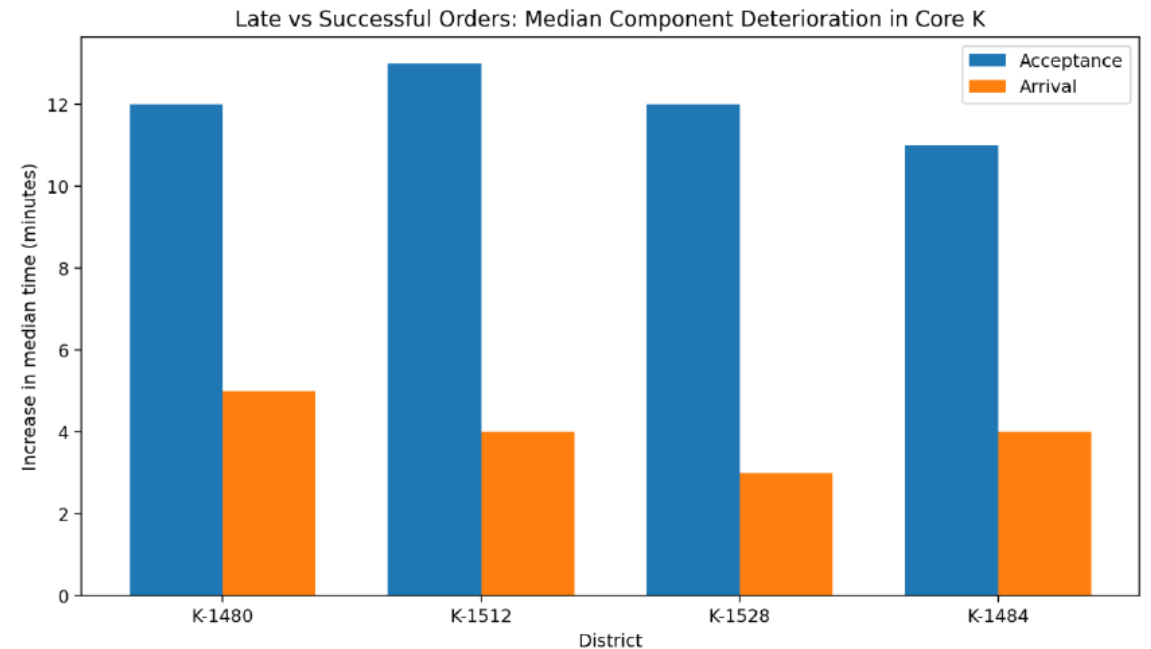
- Raw BATR15 can be misleading because districts operate under different City × Month × Weekday × Hour conditions.
- Each district was therefore compared with other districts under the same operating context, excluding itself from the benchmark.

Area	Raw BATR15 (%)	Late Orders	Context-Matched BATR15 (%)	Expected BATR15 (%)	Context Gap (pp)
U-1400	69.82	1256	69.92	81.09	-11.17
B-3728	82.88	278	83.27	90.84	-7.58
K-1528	66.38	2392	66.27	73.09	-6.81
U-1416	74.15	814	74.09	80.35	-6.26
B-3740	85.16	674	85.67	91.81	-6.15
K-1512	69.01	3367	68.43	73.73	-5.3
K-1484	67.65	1750	67.7	71.8	-4.09
K-1480	70.19	8018	70.11	73.52	-3.4
U-1428	77.32	1945	77.13	80.28	-3.15
U-1396	77.03	797	77.11	79.94	-2.83
K-1496	69.86	400	69.91	72.42	-2.51
U-1408	78.04	620	78.13	80.03	-1.9

Acceptance Time versus Arrival Time

- Late orders show a much larger increase in Acceptance Time than in Arrival Time.
- In the four main problem districts, Acceptance Median increases by 11–13 min, while Arrival rises only 3–5 min.
- These districts account for about 71% of all late orders in City K.
- The main source of delay in City K occurs before or during the Acceptance Stage, not during post-acceptance arrival.

District	Late Orders	Δ Median Acceptance	Δ Median Arrival
K-1480	8,018	+12 min	+5 min
K-1512	3,367	+13 min	+4 min
K-1528	2,392	+12 min	+3 min
K-1484	1,750	+11 min	+4 min



District-Hour Hotspots

District-level averages can mask short but severe hourly performance issues. Analysis at the District × Hour level reveals several critical hotspots in City K.

Across the major hotspots, **BATR deterioration is primarily driven by delays in the Acceptance Stage rather than post-acceptance Arrival**, making Acceptance the main operational bottleneck to investigate.

Area-Hour	Sequential Orders	BATR15 (%)	Acceptance Median Gap	Acceptance P90 Gap	Arrival P90 Gap
K-1480 H20	2846	56.11	6.0	1.0	-1.0
K-1512 H12	1258	48.97	4.0	5.0	-1.0
K-1528 H12	790	51.77	4.0	4.0	-2.0
K-1528 H13	877	57.24	7.0	4.0	-1.0

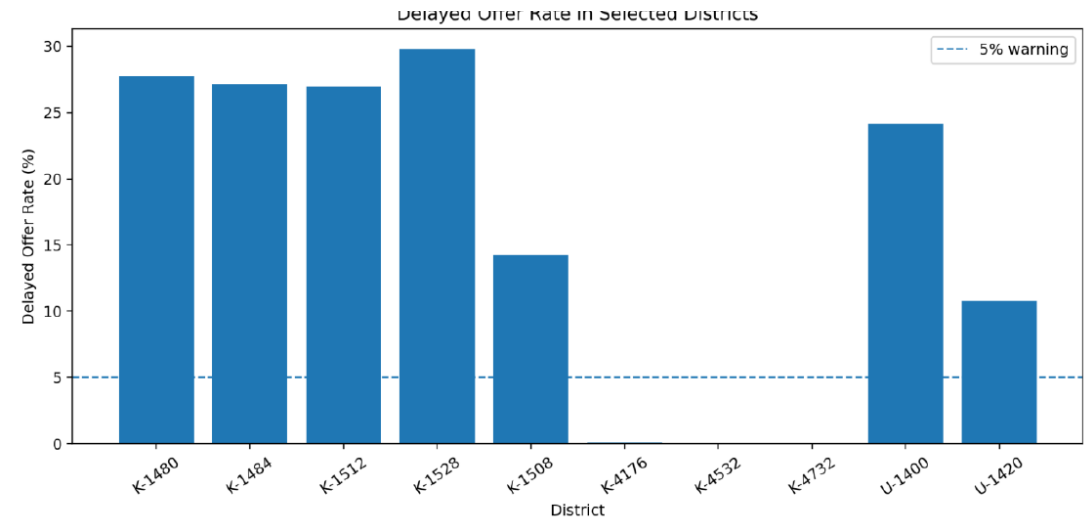
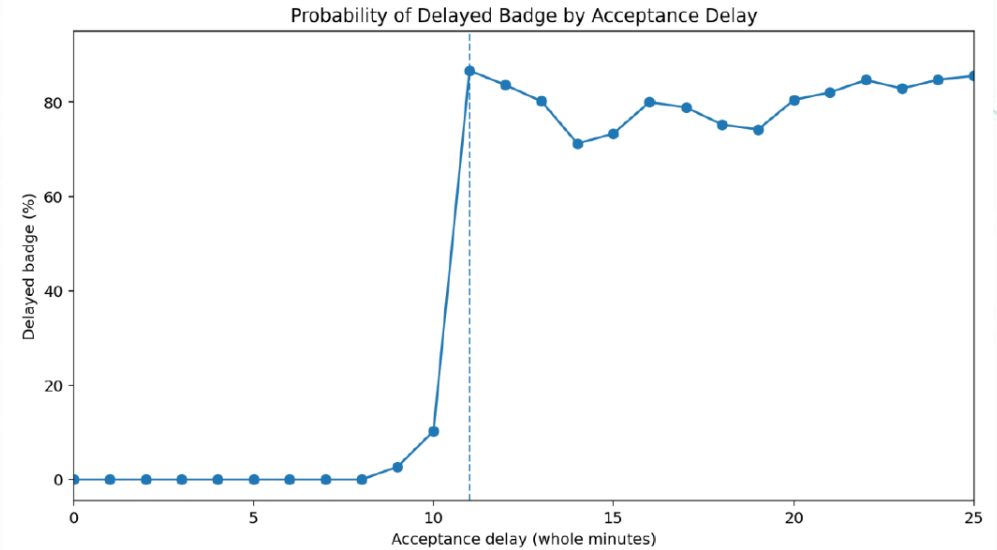
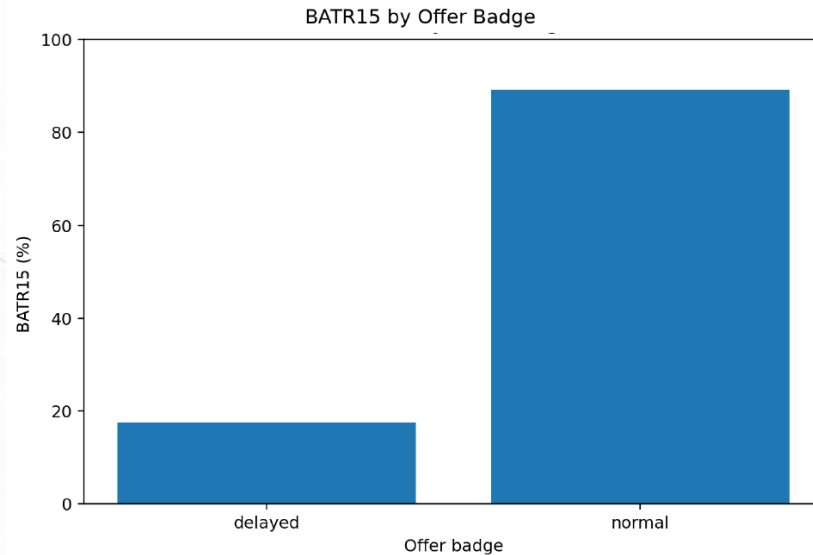
Order Load, Supply and Demand

- Higher local order load is associated with a clear drop in BATR15, especially in key City K districts.
- High local demand mainly affects the Acceptance Stage, suggesting a potential supply–demand imbalance.

District	Low-load BATR15	High-load BATR15	Change	Acceptance Change	Arrival Change
K-1480	79.48%	60.06%	-19.42 pp	+4 min	+1 min
K-1512	77.12%	60.75%	-16.37 pp	+3 min	≈ 0
K-1528	71.42%	61.03%	-10.39 pp	+3 min	≈ 0
K-1484	67.11%	63.81%	-3.31 pp	+1 min	≈ 0

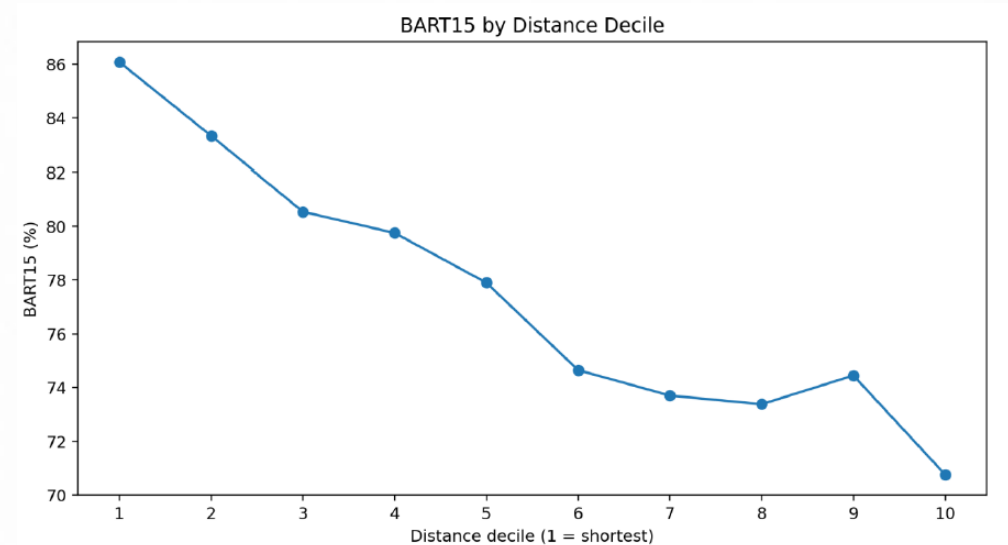
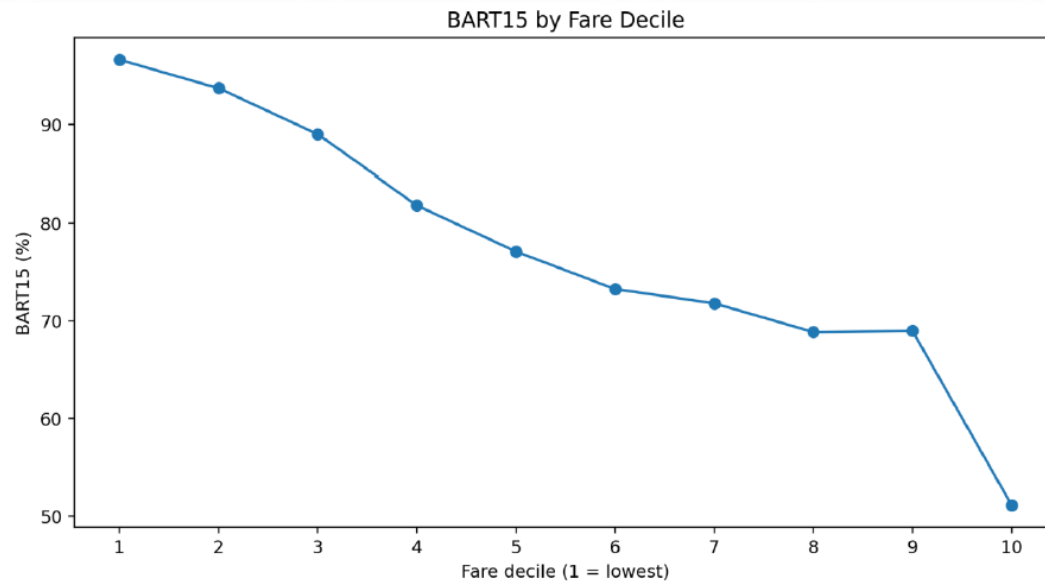
Order Offer Badge

- Delayed Badge orders have dramatically lower BATR15 (17.6% vs. 89.2% for Normal Badge).
- Although only 16.3% of orders are delayed-badge orders, they account for nearly 60% of all Late Orders.



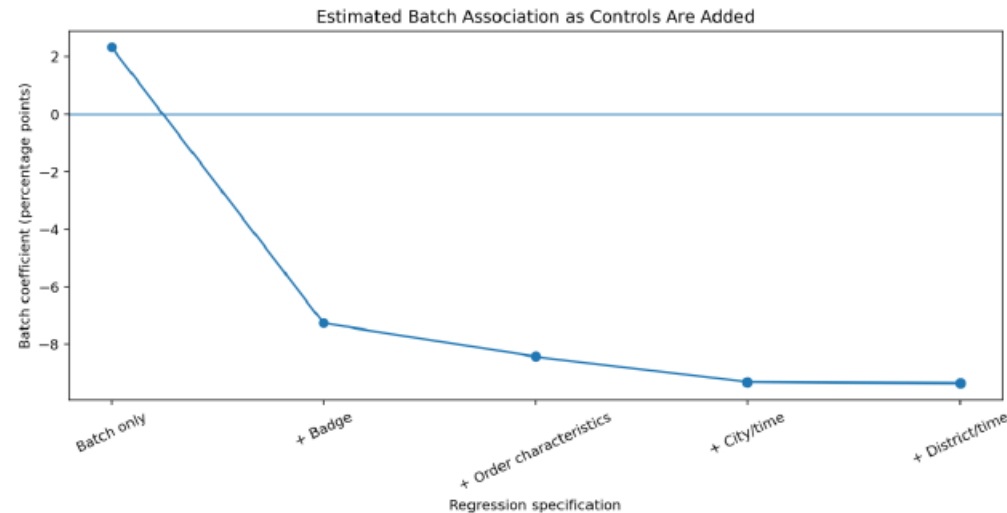
Fare, Distance and Fare per Distance

- Longer-distance orders show lower BATR15, but distance refers to restaurant-to-customer distance, not biker-to-restaurant travel.
- Higher Fare is also associated with worse BATR15, but this relationship becomes much weaker after controlling for other factors.
- Fare per Distance does not show a consistent relationship with BATR15.



Batch

- Raw results suggest Batch Orders perform slightly better (79.5% vs. 77.2% BATR15), but this is driven by selection bias.
- After matching and controlling for Badge, Distance, Fare, City and Time, Batch Orders show about 8–9 pp lower BATR15.
- The result is consistent across cities, but does not prove a causal Batch effect.



Different Problems Require Different Actions

The analysis shows that low BATR performance is not driven by a single operational issue.

- Core City K is mainly an Acceptance-side problem, especially during high-load periods.

Orders remain unaccepted for longer, while Arrival Time changes much less.

- U-1400 also has a strong Acceptance problem, but it is not explained by higher local Order Load, suggesting an issue in the Offer or Assignment process.
- B-3740, in contrast, is mainly an Arrival-side problem, where the delay occurs after the biker has already accepted the order.

The operational response should depend on where the delay is actually created.

Operational Response by Bottleneck

Bottleneck Type	What We Observe	Recommended Operational Response
Acceptance – High Load	Acceptance Time increases during peak demand, while Arrival changes much less.	Reposition available bikers from nearby areas with spare capacity and intervene before the backlog becomes severe.
Acceptance – Assignment / Offer Process	Acceptance remains weak even when local Order Load is not unusually high.	Prioritize aging orders, expand the Offer Pool earlier, and investigate Reject, Ignore, and Re-offer behavior.
Arrival – Positioning / Mobility	Orders are accepted, but bikers take too long to reach the restaurant.	Prioritize bikers with lower ETA and pre-position available bikers near high-volume Vendor clusters.

Key Principle: The intervention should match the bottleneck. A single network-wide response can increase cost or shift the problem to another area.

Different Problems Require Different Actions

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From Monitoring to Early Intervention

The dashboard provides the operational monitoring layer, while the next step is to make the process increasingly proactive.

The five core KPIs provide early signals of deterioration. For example:

- **Acceptance P90 > 8 min** indicates growing Acceptance pressure.
- **Delayed Offer Rate > 5%** can provide an early warning of deterioration in the Offer process.
- **Arrival P90 > 10 min** indicates increasing post-Acceptance travel pressure.

In addition, the predictive model can identify orders with a high probability of BATR15 failure **before the failure occurs.**

The model achieved a **ROC-AUC of 0.858** and **66.9% Late-order Recall**, suggesting that it can be used as an additional Early-Warning Layer rather than as a replacement for operational KPIs.

Recommended Next Steps

The analysis suggests moving from a single network-wide response toward **targeted, bottleneck-based operational interventions**.

I would start with three focused pilots:

- **K-1480 - Peak-Hour Supply Repositioning**
Test biker repositioning before Dinner Peak when Acceptance pressure begins to rise.
- **K-1512 / K-1528 - Earlier Offer-Pool Expansion**
Intervene before orders enter severe delay by prioritizing aging orders and expanding assignment earlier.
- **B-3740 - ETA-Based Assignment & Pre-positioning**
Prioritize better-positioned bikers and move available Supply closer to high-volume Vendor clusters.

Each pilot should be tested on a limited scale, compared with a Control or historical baseline, and expanded only when the **target KPI improves without unacceptable increases in Cost or deterioration in neighboring areas**.